

Conjecture 1

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 2603.02786__00 — generated 2026-09-16

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result would_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

Summary and provenance

Catalog id	2603.02786__00
Catalog entry	Conjecture 1
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Packing arithmetic progressions
Model verdict	proved (confidence: high)
Model's one-line claim	Fixed-size CRT-generic prime blocks, together with clustered residue classes for their multiples and a tail decomposition, give $m(n) \sim (4/3)n^{3/2}/\log n$.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2603.02786__00/output.md
Independent review	None
Model's caveats	The argument is qualitative and does not extract a useful rate of convergence.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

A proof of the $4/3$ asymptotic

All translations below are integer translations. I use interval cardinality as the packing length; using interval diameter instead changes the parameter by at most one.

Write

$$A_d(n) = \{d, 2d, \dots, \lfloor n/d \rfloor d\}, \quad x = \sqrt{n}, \quad T_n = \frac{n^{3/2}}{\log n}.$$

Theorem

$$m(n) = \left(\frac{4}{3} + o(1)\right) T_n.$$

The key new ingredient is that, for a fixed number of suitably selected primes, all pairwise CRT intersection phases can simultaneously be placed in prescribed arcs. Thus one can allow three or more convex hulls to overlap, rather than impose the path condition from the previous attempt.

The proof also handles the composite-indexed progressions explicitly. The only analytic number-theoretic input is the prime number theorem.

1. The lower bound

For completeness, I verify the prime obstruction used in the previous attempt.

Let $p, q \leq x$ be distinct primes, and suppose

$$t + A_p(n), \quad u + A_q(n)$$

are disjoint, with $u \geq t$. Put $r_p = \lfloor n/p \rfloor$. Then

$$u - t > r_p p - pq. \quad (1.1)$$

Indeed, if $0 \leq \delta = u - t \leq r_p p - pq$, choose $j \in \{1, \dots, p\}$ such that

$$\delta + jq \equiv 0 \pmod{p}.$$

Then

$$i = \frac{\delta + jq}{p}$$

satisfies $1 \leq i \leq r_p$, while $j \leq p \leq \lfloor n/q \rfloor$, because $pq \leq n$. Hence

$$t + ip = u + jq,$$

a contradiction.

Now take a packing of all $A_p(n)$, $p \leq x$, in an interval of cardinality M , and order their translation bases:

$$t_{p_1} \leq \dots \leq t_{p_s}.$$

Summing (1.1) gives

$$t_{p_s} - t_{p_1} \geq (s-1)n - \sum_{i=1}^{s-1} p_i p_{i+1} - O(n).$$

Here the rounding errors sum to $O(n)$. Also

$$\sum_{i=1}^{s-1} p_i p_{i+1} \leq \frac{1}{2} \sum_{i=1}^{s-1} (p_i^2 + p_{i+1}^2) \leq \sum_{p \leq x} p^2.$$

Translation bases of progressions contained in one interval of cardinality M differ by at most $M + x$. Consequently,

$$M \geq n\pi(x) - \sum_{p \leq x} p^2 - O(n). \quad (1.2)$$

The prime number theorem and partial summation yield

$$\pi(x) \sim \frac{x}{\log x}, \quad \sum_{p \leq x} p^2 \sim \frac{x^3}{3 \log x}.$$

Since $\log n = 2 \log x$, (1.2) implies

$$m(n) \geq \left(\frac{4}{3} - o(1)\right) T_n. \quad (1.3)$$

The rest of the proof establishes the matching upper bound.

2. Packing all multiples of one modulus

For an integer $q \geq 1$, call

$$\mathcal{B}_q(n) = \{A_{qa}(n) : 1 \leq a \leq \lfloor n/q \rfloor\}$$

the bundle associated with q .

We first give a deliberately crude but useful way to pack a bundle.

Lemma 2.1

There is an absolute constant K_0 such that, for every $N \geq 1$, the family

$$\{A_a(N) : 1 \leq a \leq N\}$$

can be partitioned into at most

$$R(N) \leq K_0 N^{3/5} \quad (2.1)$$

classes, each consisting of pairwise disjoint **unshifted** progressions.

Proof Give each $a \leq \sqrt{N}$ its own class.

For $a > \sqrt{N}$, form the intersection graph of the unshifted sets $A_a(N)$. A neighbor b of a divides some multiple $ka \leq N$. Therefore its degree is at most

$$\sum_{k \leq N/a} \tau(ka) \leq \sqrt{N} \max_{t \leq N} \tau(t).$$

The elementary divisor bound

$$\tau(t) = O(t^{1/10})$$

then gives maximum degree $O(N^{3/5})$, so greedy coloring proves (2.1).

For clarity, the divisor bound follows directly from prime factorization. For all sufficiently large primes p ,

$$e + 1 \leq 2^e \leq p^{e/10}.$$

The finitely many smaller primes contribute only an absolute multiplicative constant, since

$$\sup_{e \geq 0} \frac{e + 1}{p^{e/10}} < \infty.$$

□

Since

$$A_{qa}(n) = qA_a(\lfloor n/q \rfloor), \quad (2.2)$$

we obtain the following.

Corollary 2.2

The bundle $\mathcal{B}_q(n)$ can be packed in an interval of cardinality at most

$$(n + q) \left\lceil \frac{R(\lfloor n/q \rfloor)}{q} \right\rceil. \quad (2.3)$$

Proof Take the coloring from Lemma 2.1 and partition its colors into batches of at most q colors.

In one batch, assign a different residue $r \in \{0, \dots, q-1\}$ to each color, and translate every progression of that color by r . Different colors are disjoint modulo q ; progressions of the same color remain disjoint by (2.2). Everything in the batch lies in $[1, n+q]$.

Concatenate the batches. $\square$

For primes $p \asymp \sqrt{n}$, (2.1) gives

$$R(\lfloor n/p \rfloor) = O(n^{3/10}) = o(p). \quad (2.4)$$

The fact that only $o(p)$ residue classes are needed will let an entire bundle follow a single prime's CRT phase.

3. Simultaneous CRT phase control

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and write $\|z\|_{\mathbb{T}}$ for distance to the nearest integer.

For distinct primes p_i, p_j and residues $r_i \bmod p_i, r_j \bmod p_j$, define

$$Z_{ij}(r_i, r_j) = \frac{\text{CRT}_{p_i, p_j}(r_i, r_j)}{p_i p_j} \pmod{1}, \quad (3.1)$$

where the CRT representative is chosen in $\{0, \dots, p_i p_j - 1\}$.

Thus Z_{ij} specifies the locations, modulo $p_i p_j$, at which the two infinite residue classes intersect.

Lemma 3.1 – Generic prime blocks

Fix

$$k \geq 2, \quad 0 < a < b, \quad 0 < \eta < 1.$$

There is a constant $L = L(k, a, b, \eta)$ with the following property.

For all sufficiently large X , every set of primes in $[aX, bX]$ can be partitioned into k -element blocks and at most L exceptional primes, so that each block $p_1, \dots, p_k$ has this property:

For arbitrary open arcs $J_{ij} \subseteq \mathbb{T}$, $1 \leq i < j \leq k$, each of length η , there are residues $r_i \bmod p_i$ satisfying

$$Z_{ij}(r_i, r_j) \in J_{ij} \quad \text{for every } i < j. \quad (3.2)$$

Proof Put $E = \binom{k}{2}$. For $i \neq j$, let

$$\overline{p_j}^{(p_i)}$$

denote the inverse of p_j modulo p_i . The CRT formula gives

$$Z_{ij}(r_i, r_j) = \frac{\overline{p_j}^{(p_i)} r_i}{p_i} + \frac{\overline{p_i}^{(p_j)} r_j}{p_j} \pmod{1}. \quad (3.3)$$

Choose the r_i independently and uniformly. For an integer vector

$$h = (h_{ij})_{i < j} \in \mathbb{Z}^E,$$

writing $h_{ji} = h_{ij}$, we have

$$\mathbb{E} \exp \left(2\pi i \sum_{i < j} h_{ij} Z_{ij} \right) = \prod_{i=1}^k \mathbf{1} \left[\sum_{j \neq i} h_{ij} \overline{p_j}^{(p_i)} \equiv 0 \pmod{p_i} \right]. \quad (3.4)$$

A finite Fourier criterion There is an integer $H = H(k, \eta)$ such that (3.2) holds whenever the expectation in (3.4) vanishes for every nonzero h with

$$\|h\|_\infty \leq H. \quad (3.5)$$

To see this, choose a nonnegative smooth periodic function f , supported inside an arc of length η , with

$$\mu = \int_{\mathbb{T}} f > 0.$$

For prescribed arc centers c_{ij} , consider

$$F(z) = \prod_{i < j} f(z_{ij} - c_{ij}).$$

Its constant Fourier coefficient is μ^E . Its Fourier series is absolutely convergent, and the absolute sum of coefficients outside $\|h\|_\infty \leq H$ tends to zero as $H \rightarrow \infty$, uniformly in the centers c_{ij} .

Choose H so that this tail is less than $\mu^E/2$. Under (3.5), (3.4) then gives

$$\mathbb{E}F(Z) \geq \mu^E/2 > 0.$$

Some residue vector therefore satisfies (3.2).

There are few bad tuples Call an ordered tuple of distinct primes bad if some nonzero h satisfying (3.5) makes all the congruences in (3.4) hold.

Fix such an h , and choose i, j with $h_{ij} \neq 0$. Fix all primes in the tuple except p_j . The congruence at i has the form

$$h_{ij} \overline{p_j}^{(p_i)} \equiv -c \pmod{p_i}, \quad (3.6)$$

where c is fixed.

For $p_i > H$, if $c = 0$, (3.6) has no solution. Otherwise it forces p_j into a single residue class modulo p_i . Since

$$p_i \geq aX, \quad p_j \in [aX, bX],$$

there are only $O_{a,b}(1)$ candidate integers for p_j .

Consequently, in any set of M primes from this interval, the number of bad ordered k -tuples is

$$O_{k,a,b,\eta}(M^{k-1}).$$

For M larger than a suitable constant, this is less than the number $(M)_k$ of ordered distinct tuples. Hence a good block exists.

Repeatedly remove good blocks until only a bounded number of primes remain. $\square$

A useful point is that this counting argument does **not** require information about primes in arithmetic progressions with growing moduli. Once the other primes are fixed, there are only boundedly many candidate integers for the remaining prime.

4. Keeping many residue classes close in CRT phase

We need a simple simultaneous-approximation observation.

Lemma 4.1 — A large centered Bohr set

Fix $d \geq 1$ and $\rho > 0$. There is $c = c(d, \rho) > 0$ such that, for every integer $q \geq 2$ and every $c_1, \dots, c_d \in \mathbb{Z}$, the set

$$\left\{ s \bmod q : \left\| \frac{c_j s}{q} \right\|_{\mathbb{T}} < \rho \text{ for all } j \right\}$$

has at least cq elements.

Proof Choose an integer $M > 1/\rho$. Partition $[0, 1)^d$ into M^d congruent boxes and consider the q points

$$\left(\frac{c_1 s}{q}, \dots, \frac{c_d s}{q} \right) \pmod{1}, \quad s \bmod q.$$

One box contains at least q/M^d residues. Fix one of them, s_0 , and subtract s_0 from all the others. The resulting distinct residues satisfy all the required inequalities. $\square$

Applied with $q = p_i$ and coefficients $\overline{p_j}^{(p_i)}$, this gives linearly many residue perturbations whose effects on every CRT phase involving p_i are small.

5. Packing the main prime range

Fix constants

$$0 < \epsilon < 1 < C.$$

Let $\mathcal{G}_{\epsilon, C}(n)$ be the family of all progressions $A_d(n)$ whose index has a prime divisor in

$$[\epsilon x, Cx].$$

I will prove

$$\limsup_{n \rightarrow \infty} \frac{m(\mathcal{G}_{\epsilon, C}(n))}{T_n} \leq 2 \int_{\epsilon}^C (1 - u^2)_+ du. \quad (5.1)$$

When bundles overlap, we may pack labelled copies first and then delete duplicates. Thus it is enough to pack all the bundles $\mathcal{B}_p(n)$ in this prime range.

5.1 Packing one good block

Fix an integer $k \geq 2$ and

$$0 < \nu < \epsilon^2/4.$$

Put

$$h(u) = (1 - u^2)_+, \quad \rho = \frac{\nu}{100C^2}.$$

Consider a block of primes

$$p_1, \dots, p_k \in [ux, vx] \subseteq [\epsilon x, Cx]$$

having the property in Lemma 3.1 for arcs of length ρ .

Set

$$g = \lceil n(h(u) + \nu) \rceil, \quad W = n + \lceil Cx \rceil + 1, \quad S_i = (i - 1)g.$$

We will place the whole bundle associated with p_i inside

$$I_i = [S_i, S_i + W - 1].$$

For $i < j$, if $I_i \cap I_j \neq \emptyset$, its real length is at most $W - g$. For all sufficiently large n ,

$$\begin{aligned} W - g &\leq (\min(u^2, 1) - \nu/2)n \\ &\leq p_i p_j - \nu n/2. \end{aligned} \quad (5.2)$$

Also $p_i p_j \leq C^2 n$.

Project $I_i \cap I_j$ into $\mathbb{T}$ by dividing by $p_i p_j$. By (5.2), the complement of this forbidden arc has length at least

$$\frac{\nu}{2C^2} = 50\rho.$$

Choose an open arc J_{ij} of length ρ in the complement, at circular distance at least 4ρ from the forbidden arc. If the intervals are disjoint, choose any arc.

Lemma 3.1 supplies residues $r_i \bmod p_i$ such that

$$Z_{ij}(r_i, r_j) \in J_{ij} \quad (i < j). \quad (5.3)$$

Now put

$$N_i = \lfloor n/p_i \rfloor$$

and color the unshifted $A_a(N_i)$ using Lemma 2.1. There are

$$R(N_i) = O_\epsilon(n^{3/10}) = o(p_i) \quad (5.4)$$

colors.

For each i , Lemma 4.1 supplies at least $c_{k,\rho} p_i$ residues $s \bmod p_i$ satisfying

$$\left\| \frac{\overline{p_j^{(p_i)}} s}{p_i} \right\|_{\mathbb{T}} < \rho \quad (j \neq i). \quad (5.5)$$

By (5.4), there are enough distinct such residues to assign one, denoted $s_{i,c}$, to every color c .

For color c , choose $t_{i,c} \in \{0, \dots, p_i - 1\}$ such that

$$S_i + t_{i,c} \equiv r_i + s_{i,c} \pmod{p_i}.$$

Translate every $A_{ap_i}(n)$ of that color by $S_i + t_{i,c}$.

These translates lie in I_i . Within one bundle:

- progressions of the same color are disjoint by their unshifted coloring;
- different colors occupy distinct residues modulo p_i .

For two different bundles, their relevant CRT phase is

$$\begin{aligned} Z_{ij}(r_i + s_{i,c}, r_j + s_{j,c'}) &= Z_{ij}(r_i, r_j) \\ &\quad + \frac{\overline{p_j^{(p_i)}} s_{i,c}}{p_i} + \frac{\overline{p_i^{(p_j)}} s_{j,c'}}{p_j} \pmod{1}. \end{aligned} \quad (5.6)$$

By (5.5), this differs from the phase in (5.3) by circular distance less than 2ρ . It therefore remains outside the forbidden projection of $I_i \cap I_j$.

A common point of two progressions would belong to both envelopes and have precisely the CRT phase in (5.6), which is impossible.

We have proved that all bundles in this block can be packed in an interval of cardinality at most

$$(k-1)g + W. \quad (5.7)$$

5.2 Summing the blocks

Partition $[\epsilon, C]$ into finitely many intervals

$$\epsilon = u_0 < u_1 < \dots < u_J = C.$$

For each bin, apply Lemma 3.1 to its primes, obtaining k -element good blocks and $O(1)$ exceptional primes.

An exceptional prime's whole bundle fits in an interval of cardinality W , since (5.4) eventually gives $R(N_i) \leq p_i$. Thus all exceptions cost $O(n) = o(T_n)$.

Let

$$M_j = \#\{p : u_{j-1}x \leq p < u_jx\}, \quad g_j = \lceil n(h(u_{j-1}) + \nu) \rceil.$$

Using (5.7) and concatenating blocks,

$$m(\mathcal{G}_{\epsilon, C}(n)) \leq \sum_{j=1}^J M_j \left(g_j + \frac{W}{k} \right) + O(n). \quad (5.8)$$

The prime number theorem gives

$$M_j = (u_j - u_{j-1} + o(1)) \frac{x}{\log x}.$$

Dividing (5.8) by T_n , we get

$$\limsup_{n \rightarrow \infty} \frac{m(\mathcal{G}_{\epsilon, C}(n))}{T_n} \leq 2 \sum_{j=1}^J (u_j - u_{j-1}) \left(h(u_{j-1}) + \nu + \frac{1}{k} \right).$$

All parameters here are fixed before taking $n \rightarrow \infty$. We may therefore let the partition mesh tend to zero, then let $\nu \downarrow 0$ and $k \rightarrow \infty$. This proves (5.1).

For $C > 1$, the resulting bound is

$$m(\mathcal{G}_{\epsilon, C}(n)) \leq \left(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3 + o(1) \right) T_n. \quad (5.9)$$

6. Two tools for the large-prime tail

Lemma 6.1 – Folding a scaled packing

Suppose a family of finite integer sets, each of diameter at most N , has a packing in an interval of cardinality M . For an integer $a \geq 1$, the family obtained by multiplying every set by a has a packing of cardinality at most

$$M + aN + a. \quad (6.1)$$

Proof Translate the given packing into $[0, M - 1]$, and put

$$H = \lceil M/a \rceil.$$

Assign a packed set B to the strip indexed by $j \in \{0, \dots, a - 1\}$ containing its minimum:

$$jH \leq \min B < (j + 1)H.$$

Replace it by

$$a(B - jH) + j.$$

Sets from the same strip remain disjoint. Sets from different strips have distinct residues modulo a . All resulting sets lie in an interval of cardinality at most

$$aH + aN \leq M + aN + a.$$

□

In particular, because

$$A_{ap}(n) = aA_p(\lfloor n/a \rfloor), \quad (6.2)$$

folding lets us scale a prime-indexed packing without multiplying its leading packing length by a .

Lemma 6.2 — Packing large prime indices

There is an absolute constant B such that, whenever $P \geq 2\sqrt{N}$,

$$m_{\{p \text{ prime} : P < p \leq N\}}(N) \leq B \left(\frac{N^2}{P \log P} + N \log(2N) \right). \quad (6.3)$$

If $P > N$, the family is empty.

Proof Use dyadic bins $(Q, 2Q]$, starting with $Q = P$. In one such bin, put

$$K = \left\lfloor \frac{Q^2}{2N} \right\rfloor.$$

Since $Q \geq 2\sqrt{N}$,

$$K \geq \frac{Q^2}{4N}. \quad (6.4)$$

Any group of at most K prime-indexed progressions from the bin can be packed inside $[1, 2N]$.

To see this, insert them successively. For a new prime p , choose a shift $r \in \{0, \dots, p-1\}$. Each previously occupied point forbids at most one residue $r \bmod p$. The previous progressions contain fewer than

$$K \frac{N}{Q} \leq \frac{Q}{2} < p$$

points, so some residue remains available. The resulting progression lies in $[1, 2N]$.

By (6.4), the total length for a bin is at most

$$2N \left(\frac{4N\pi(2Q)}{Q^2} + 1 \right).$$

Using $\pi(t) = O(t/\log t)$ and summing over dyadic Q ,

$$\sum_{Q=P, 2P, \dots} \frac{1}{Q \log Q} \leq \frac{2}{P \log P}.$$

There are $O(\log(2N))$ bins. These estimates give (6.3). $\square$

7. Covering every index

Fix $0 < \epsilon < 1$ and $C > 2$, and set

$$Y = \lceil n^{1/5} \rceil.$$

For large n ,

$$Y^2 < \epsilon x.$$

We cover all indices $d \leq n$ by four types of families:

1. the main family $\mathcal{G}_{\epsilon, C}(n)$;
2. bundles $\mathcal{B}_q(n)$ for

$$q \in \mathcal{Q}_n := \{q \in \mathbb{Z} : Y \leq q \leq Y^2\} \cup \{p \text{ prime} : Y \leq p < \epsilon x\};$$

3. individual progressions with $d < Y$;
4. for each $1 \leq a < Y$, the tail family

$$\{A_{ap}(n) : p \text{ prime}, p > Cx, ap \leq n\}.$$

Here is a verification that this is a cover.

If d has a prime divisor in $[Y, Cx]$, it is covered by type 1 or type 2.

Otherwise, every prime factor of d is either smaller than Y or larger than Cx . There is at most one factor larger than Cx , including multiplicity, because $C > 1$ and $d \leq n$. Hence

$$d = s \quad \text{or} \quad d = sp,$$

where all prime factors of s are smaller than Y , and $p > Cx$ is prime when present.

If $s \geq Y$, multiply its prime factors until the product first reaches Y . This gives a divisor

$$Y \leq q < Y^2,$$

so type 2 applies. If $s < Y$, then either $d < Y$, giving type 3, or $d = sp$, giving type 4.

Thus no indices are omitted.

7.1 Cost of the small-divisor bundles

By Corollary 2.2 and Lemma 2.1,

$$\begin{aligned} M_Q &\leq (n + \epsilon x + 1) \sum_{q \in Q_n} (1 + K_0 n^{3/5} q^{-8/5}) \\ &\leq (n + \epsilon x + 1) (\pi(\epsilon x) + Y^2 + O(n^{3/5} Y^{-3/5})). \end{aligned} \quad (7.1)$$

Now

$$Y^2 = O(n^{2/5}), \quad n^{3/5} Y^{-3/5} = O(n^{12/25}).$$

Both give $o(T_n)$ after multiplication by $n + O(x)$, since $37/25 < 3/2$. The prime number theorem therefore yields

$$M_Q \leq (2\epsilon + o(1))T_n. \quad (7.2)$$

The individual progressions with $d < Y$, packed separately, cost at most

$$nY = o(T_n). \quad (7.3)$$

7.2 Cost of the large-prime tail

Put

$$P = Cx, \quad N_a = \lfloor n/a \rfloor.$$

For every $a < Y$, we have $P \geq 2\sqrt{N_a}$. By Lemma 6.2, followed by folding and (6.2), the a -th tail family can be packed in length at most

$$B \left(\frac{N_a^2}{P \log P} + N_a \log(2N_a) \right) + n + a.$$

Summing over $a < Y$, and using

$$\sum_{a \geq 1} a^{-2} < 2, \quad \sum_{a < Y} a^{-1} = O(\log Y),$$

gives

$$M_{\text{tail}} = O \left(\frac{n^2}{P \log P} \right) + O(n \log^2 n + nY + Y^2). \quad (7.4)$$

The second term is $o(T_n)$. Since

$$\frac{n^2}{Cx \log(Cx)} \leq \frac{2}{C} T_n,$$

there is an absolute constant B_0 , independent of ϵ, C , such that

$$M_{\text{tail}} \leq \left(\frac{B_0}{C} + o(1) \right) T_n. \quad (7.5)$$

8. Completing the upper bound

Concatenate the packings of the covering families, then delete any duplicate copies.

Combining (5.9), (7.2), (7.3), and (7.5),

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} &\leq \left(\frac{4}{3} - 2\epsilon + \frac{2}{3}\epsilon^3 \right) + 2\epsilon + \frac{B_0}{C} \\ &= \frac{4}{3} + \frac{2}{3}\epsilon^3 + \frac{B_0}{C}. \end{aligned}$$

This holds for every fixed $0 < \epsilon < 1$ and $C > 2$. Letting $\epsilon \downarrow 0$ and $C \rightarrow \infty$,

$$\limsup_{n \rightarrow \infty} \frac{m(n)}{T_n} \leq \frac{4}{3}.$$

Together with (1.3),

$$m(n) \sim \frac{4}{3} \frac{n^{3/2}}{\log n}.$$

What closes the earlier gaps

Two points distinguish this argument from the previous path-based attempt:

- **Multiple overlapping hulls:** Lemma 3.1 simultaneously controls every pairwise CRT phase in a fixed block. No triple-overlap restriction is imposed.
- **Composite indices:** Lemma 2.1 places an entire prime bundle in $o(p)$ residue classes. Lemma 4.1 lets all these classes remain close to the bundle's chosen CRT phase. The indices not captured by main-range bundles are explicitly covered and bounded in Section 7.

All block sizes and phase tolerances are fixed before $n \rightarrow \infty$. This order of limits is sufficient for the asymptotic, although it produces no practical convergence rate.